
Map
Ask Haze
When Haze Stop
Real-Time Detection, Forecast, and Simulation for Air Pollution
George Town
4 km haze
Medan
4 km haze
2 km
3.5 km haze
1.5 km
Singapore
km haze
2.3 km
2.3 km smoke
250 m
7 km
3 km
Kuching
4 km haze
1.7 km
Air Pollutant Index (API)
0 50 100 150 200 300
Above 100 is unhealthy. Schools close above 180.
Squares are airports: darker, the shorter the observer could see. A red ring is reported smoke.
Haze
Stations
Fires
Wind
Visibility
Jan 2026
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
History
Live
Now
Sun 11 Oct, 06:00
Now
Tonight
Tomorrow
Day after
Mon 6
Tue 7
Wed 7
Thu 8
Fri 9
Sat 10
Sun 11
Mon 12
Tue 13
Wed 14

PROBLEM

# The haze is back, and the air is unhealthy to breathe.

## Malaysia closes schools in several states, territories due to haze

By **Reuters**

October 9, 2026 10:16 AM GMT+8 · Updated October 9, 2026



**Toxic smog shuts schools in Malaysia as pollution hits record levels in Singapore**

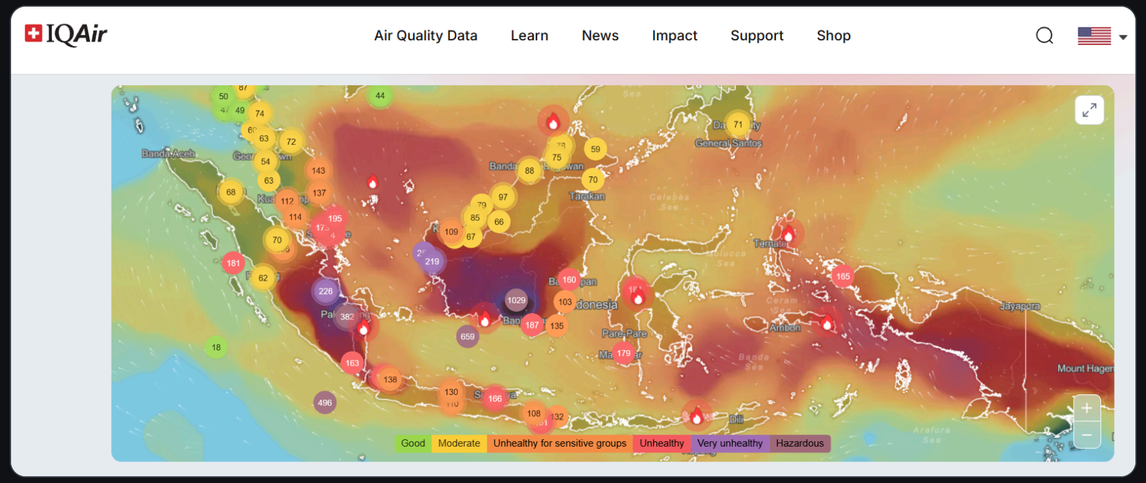
9 October 2026

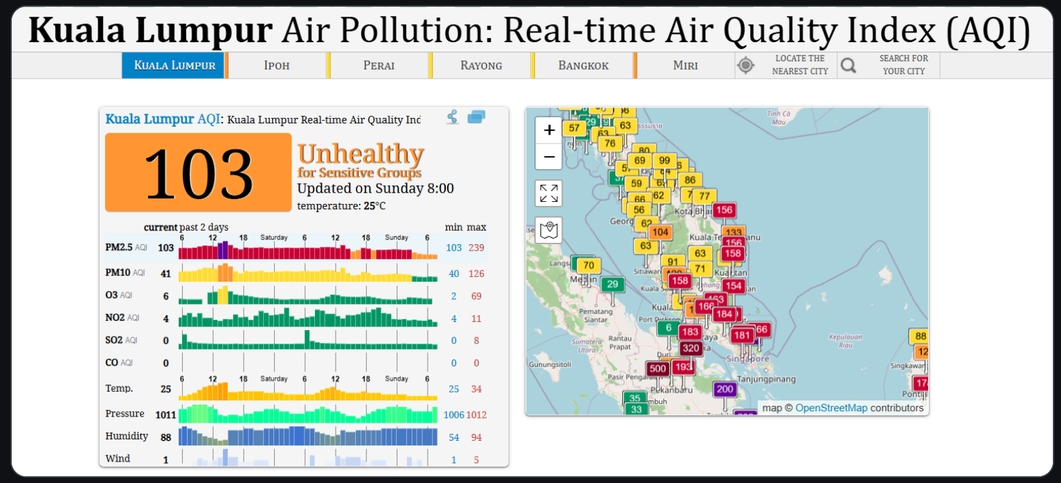




PROBLEM

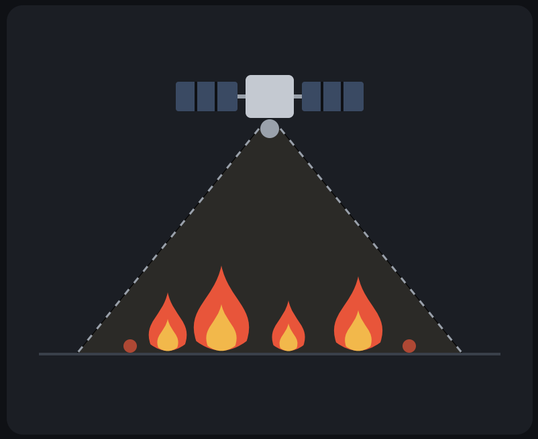
# Informational sources for haze data are limited.



HOW IT WORKS

# Four public data sources, none more than three hours old.

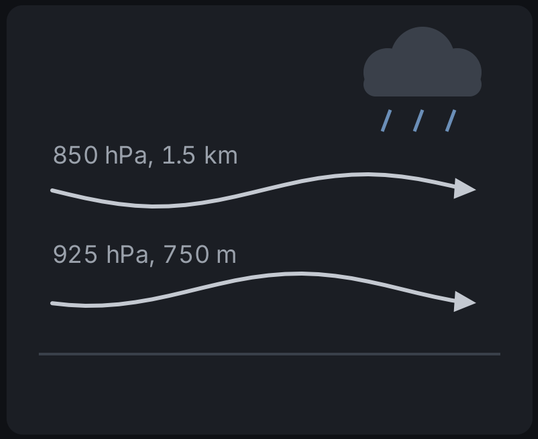


## Fire hotspots

NASA FIRMS, 3 satellites

Where fires burn and how hot. Gas flares and industry screened out.

Every 3 hours

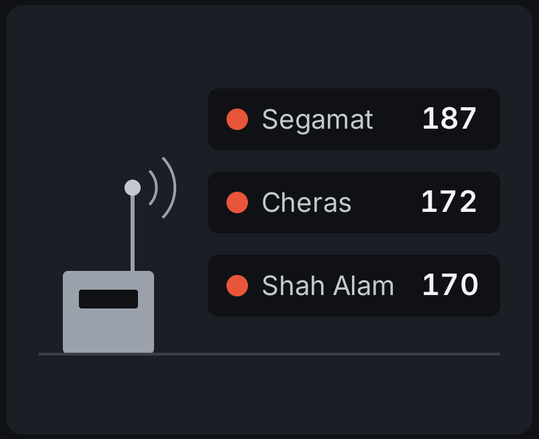



## Wind and rain

Open-Meteo forecast

Wind at the height smoke travels, and the rain that washes it out, 84 hours ahead.

Every hour

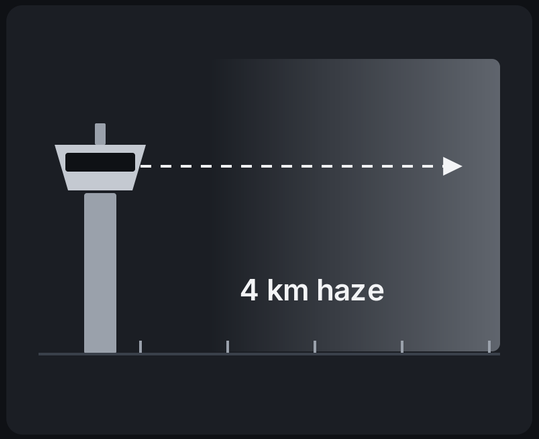



## Air quality

DOE Malaysia, 68 stations

The Air Pollutant Index that school and work rules read.

Every hour




## Airport visibility

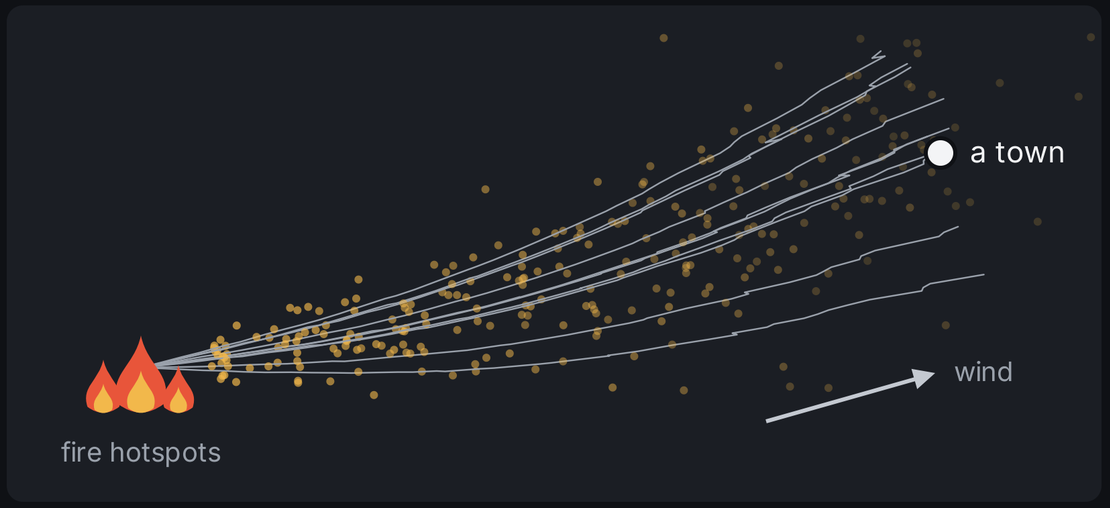
METAR reports, 29 airports

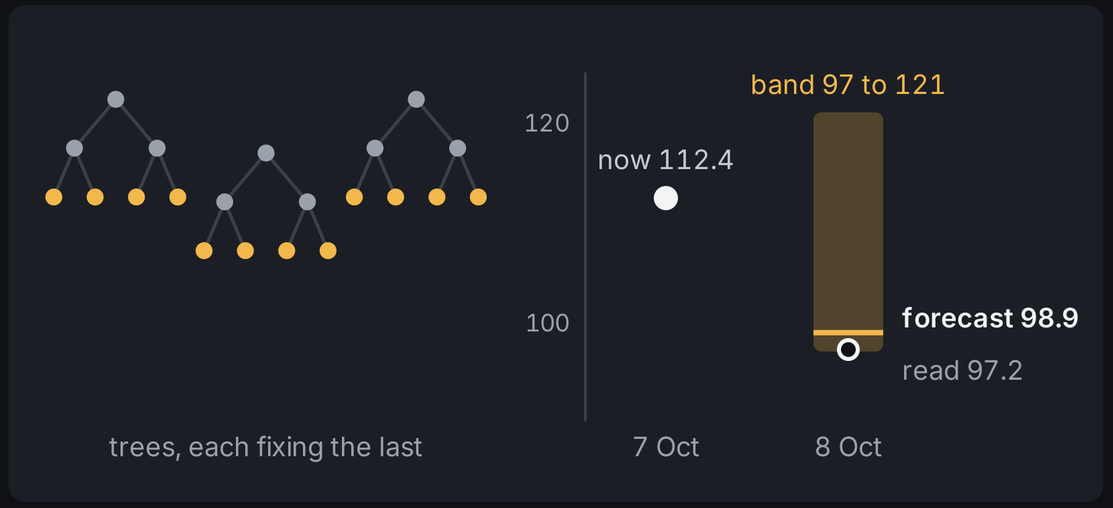
How far the observer could see, and whether they called it haze or smoke.

Every hour

MODEL

# Two Engine Aggregation for Detection




## Simulator Engine Monte Carlo smoke simulation

- In: fire hotspots, wind and rain
- Particles from every fire, carried by the wind
- Ten randomised runs, 84 hours ahead

## Forecast Engine LightGBM

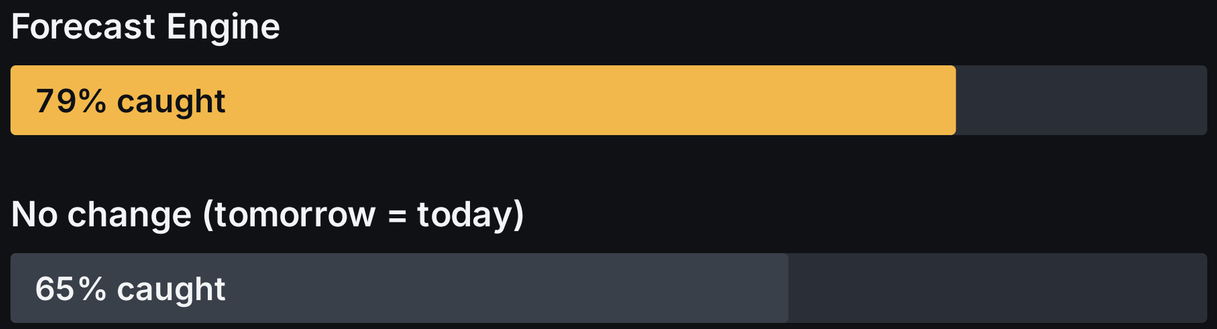
- In: the air now, simulated smoke, fires, wind
- Trained on 189 instruments since July 2025
- Out: peak PM2.5 for two days, with a range

PERFORMANCE

# Catches unhealthy air a day ahead.

79%

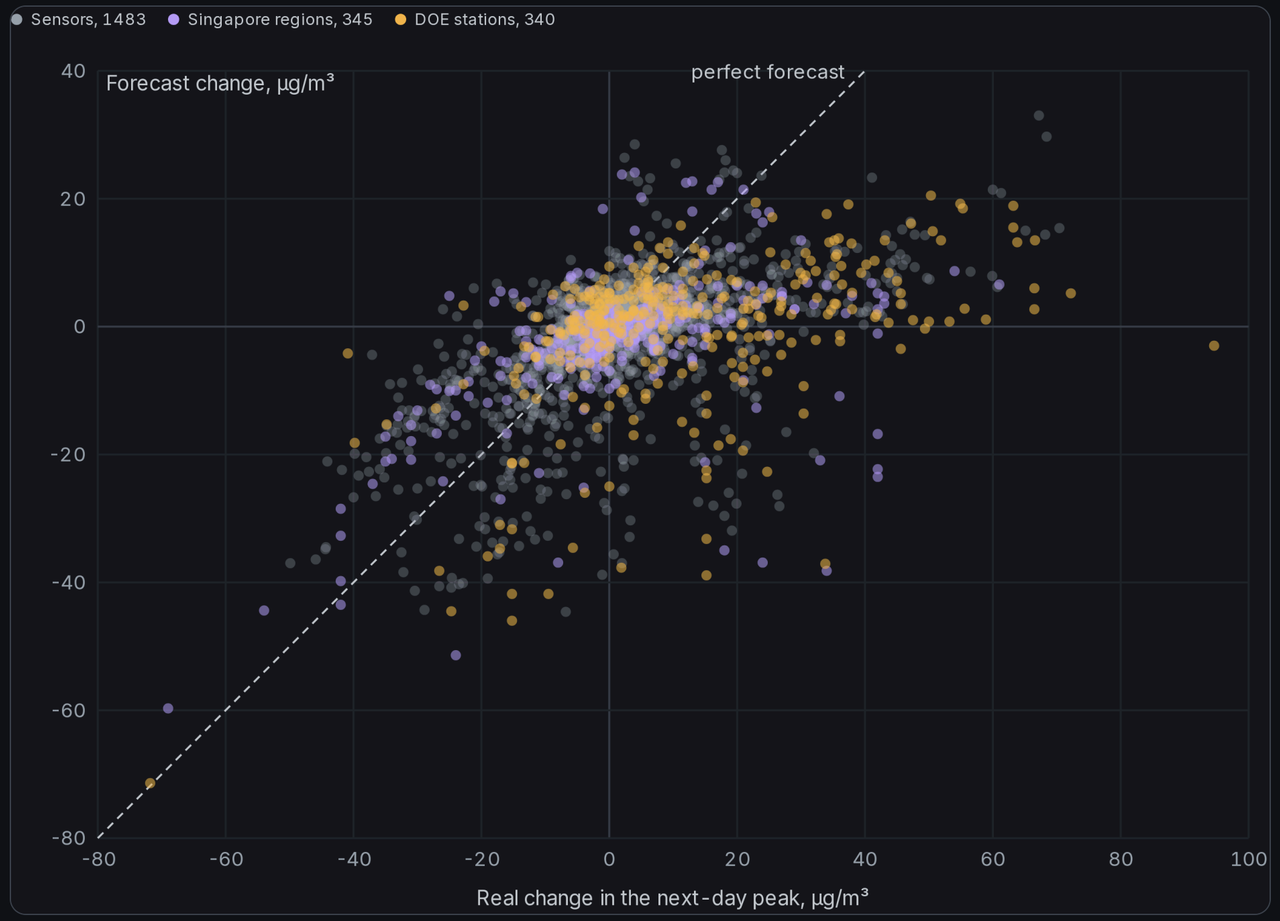
of days above API 150 at DOE stations caught a day ahead.



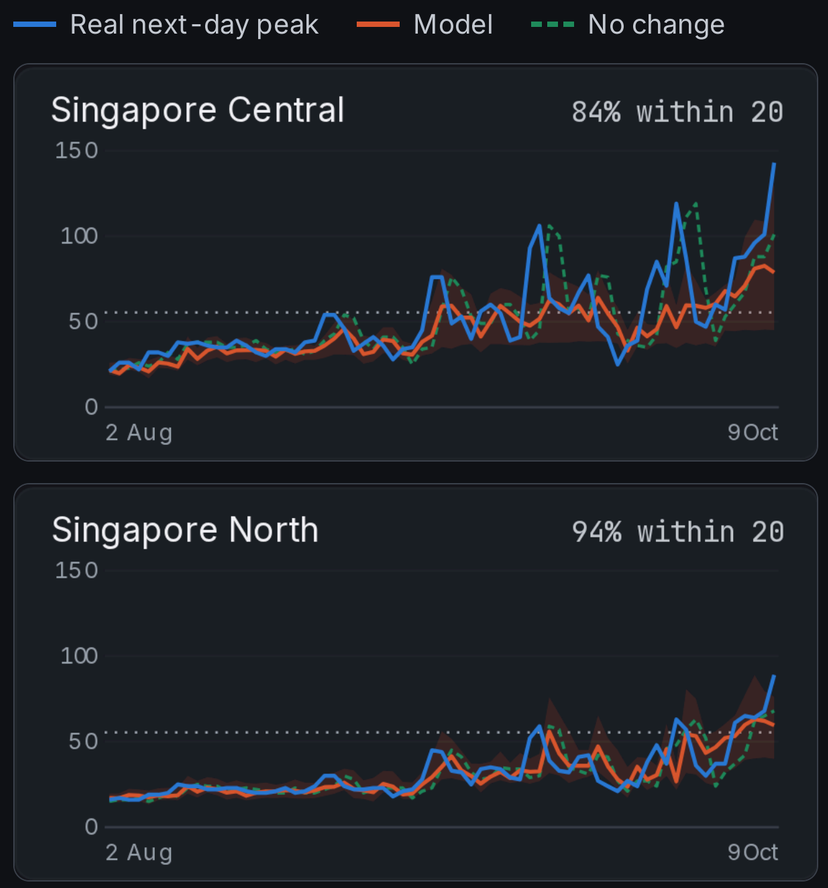

156 such days across 340 station-days, 3 to 9 October 2026.
False alarms: 7%.

PERFORMANCE

# The forecast follows what the stations read.




Each dot is one station-day. On the diagonal, the forecast change equals the real one.




All five Singapore regions: 84 to 94% of days within 20 µg/m³.

# Demo